A Gödel Modal Logic Over Witnessed Models

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We introduce **GW**, a Gödel modal logic based on Kripke models in which the value of each modal formula is witnessed by an accessible world. This witnessed semantics eliminates the limit-based phenomena that preclude the finite model property in the usual Kripke semantics for Gödel modal logics, thereby yielding a more constructive semantic framework. We present a sound and complete refutation calculus for **GW** and design a terminating backward proof-search procedure with counter-model generation. As a direct consequence of this procedure, **GW** enjoys the finite model property.

1 Introduction

Gödel modal logics, **GK**s, provide a natural framework for reasoning about graded notions of incompleteness, uncertainty, and vagueness by means of necessity and possibility operators evaluated over truth degrees. When interpreted over the real unit interval, Gödel logic captures a well-behaved notion of implication based on residuation, which makes it particularly suitable for applications in knowledge representation, as well as in multi-agent and distributed systems.

In general, the semantics of **GK** is based on Gödel Kripke models (G-*models*), a natural generalization of classical Kripke semantics in which both the valuation of propositions at each world and the accessibility relation take values in the standard Gödel algebra $[0,1]$, i.e. the algebraic semantics of the intermediate Gödel–Dummett Logic. The minimal logics over G-models were thoroughly investigated by Caicedo and Rodríguez in [5, 6]. In this setting, the interpretation of the modal operators is defined by means of infima and suprema computed over (possibly infinite) sets of accessible worlds. However, these bounds need not be attained: the infimum may be strictly smaller than every truth value in the set, and dually for the supremum. Consequently, the truth value of a modal formula may depend on limit constructions rather than on concrete worlds of the model. This phenomenon has significant logical consequences. For instance, [5] shows that the formula $\Box\neg\neg\varphi \to \neg\neg\Box\varphi$ is valid in all finite Gödel Kripke models, but fails in a suitable infinite model (see Fig. 1). Hence, the underlying logic does not enjoy the finite model property, and semantic arguments may require genuinely infinitary behaviour.

Witnessed models provide a principled way to overcome these difficulties. By requiring that the relevant infima and suprema be realized by actual accessible worlds, witnessed semantics ensures that existential and universal behaviour is supported by explicit witnesses. In this way, the evaluation of modal operators more closely mirrors the behaviour of guarded quantification in first-order logic and restores a constructive reading of the semantics. Each modal statement is thus justified by concrete worlds rather than by purely asymptotic approximations.

M. Bílková, M. Gattinger, I. van der Giessen, M. Girlando, Y. Wang (Eds.)
Advances in Modal Logic 2026 (AiML 2026)
EPTCS 447, 2026, pp. 355–373, doi:10.4204/EPTCS.447.20

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The notion of witnessed models has appeared in several related contexts. For example, in [7] an axiomatization is given for witnessed semantics in fuzzy predicate logics. Since quantified statements can often be translated into modal ones, this connection is conceptually well grounded. Along these lines, Hájek considers in [13] the axioms $\Diamond(\varphi \to \Box\varphi)$ and $\Diamond(\Diamond\varphi \to \varphi)$, and proves that the resulting logic $S5(G)^w$ is strongly complete with respect to witnessed universal Kripke models.

Witnessed semantics has also proved fruitful in applied settings. In the context of Fuzzy Description Logics, [4] studies a system based on witnessed Gödel semantics and shows that it enjoys the finite model property and a decidable satisfiability problem with exponential-time complexity. More recently, [18] provides an axiomatization for Gödel modal logic interpreted over witnessed Kripke models.

In general, witnessed models tend to ensure compactness-like properties and often yield the finite model property and decidability results. These features are crucial for automated reasoning, since they allow infinitary semantic arguments to be replaced by finite combinatorial ones. For these reasons, studying Gödel modal logics over witnessed models not only clarifies their semantic foundations, but also strengthens their connections with well-behaved fragments of first-order logic that play a central role in computer science, thereby enhancing both their theoretical robustness and their applicability.

From a proof-theoretic perspective, witnessed semantics aligns more closely with syntactic calculi. The existence of witnesses enables canonical model constructions that are finite or finitely generated, supports filtration techniques, and allows completeness proofs to proceed by standard algebraic or canonical methods. Therefore, establishing a sound and complete calculus for Gödel modal logics over witnessed models is not only feasible but also conceptually justified, providing a proof-theoretic counterpart to their improved semantic behaviour and paving the way for effective automated reasoning procedures.

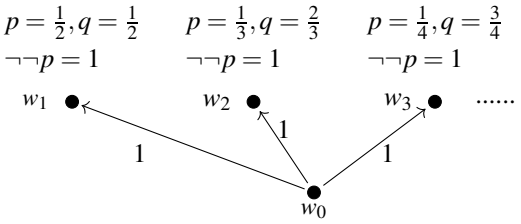
In line with the above considerations, in [11] we introduced the logic $\mathbf{GW}^{\mathrm{c}}$ characterized by witnessed crisp Gödel Kripke models, where "crisp" means that the accessibility relation takes values in the set $\{0,1\}$. For this logic, we developed a constraint-based refutation calculus inspired by the tableau system of [3], together with a terminating proof-search procedure that yields finite countermodels whenever proof search fails. As a consequence, we established that $\mathbf{GW}^{\mathrm{c}}$ enjoys the finite model property. In the present work, we extend these results to the logic **GW**, characterized by witnessed Gödel Kripke models without the crispness restriction. Our development follows the strategy adopted in [11]. We introduce a constraint-based refutation calculus for **GW**, called $\mathscr{C}_{\mathrm{GW}}$, and prove its soundness and completeness with respect to witnessed models. Moreover, we design a terminating proof-search procedure for $\mathscr{C}_{\mathrm{GW}}$ based on a standard backward search strategy. Notably, when proof search fails, the computation can be traced to construct a finite countermodel for the given formula, and this proves the finite model property for **GW**. Finally, we provide an implementation of the procedure, called `gwref` (available at [10]).

2 Basic Definitions

Formulas are built over a countable set of propositional variables $\mathscr{V}$, using the propositional connectives $\wedge, \vee, \to$, the constant $\bot$ and the modal connectives $\Box, \Diamond$. We introduce the following abbreviations: $\neg\varphi$ stands for $\varphi \to \bot$; $\varphi_1 \leftrightarrow \varphi_2$ stands for $(\varphi_1 \to \varphi_2) \wedge (\varphi_2 \to \varphi_1)$. The symbols $\wedge$, $\vee$ and $\to$ are also used to denote algebraic operations. The *Gödel algebra* is a structure $\langle [0,1], \wedge, \vee, \to \rangle$ where:

$$a \wedge b = \min(a,b) \qquad a \vee b = \max(a,b) \qquad a \to b = \begin{cases} 1 & \text{if } a \le b \\ b & \text{if } a > b \end{cases}$$

A G-*model* (*Gödel model*) $\mathfrak{M}$ is a structure $\langle W, R, e \rangle$ where W is a nonempty set (the set of worlds), R is a map $W \times W \to [0,1]$ (the accessibility relation) and $e : W \times \mathscr{V} \to [0,1]$ (the evaluation function). If



$W = \{ w_j \mid j \geq 0 \}$
$R(w_0, w_0) = 0 \qquad e(w_0, p) = e(w_0, q) = 0$
$R(w_0, w_k) = 1 \;\; k \geq 1 \qquad e(w_k, p) = \frac{1}{k+1} \;\; k \geq 1$
$e(w_k, q) = \frac{k}{k+1} \;\; k \geq 1$

$$e(w_0, \Box p) \;=\; \inf_{j\geq 0}\{ R(w_0,w_j) \to e(w_j,p) \} \;=\; \inf\{ \overbrace{0 \to 0}^{1}, \overbrace{1 \to (1/2)}^{1/2}, \overbrace{1 \to (1/3)}^{1/3}, \overbrace{1 \to (1/4)}^{1/4}, \dots \} \;=\; 0$$

$$e(w_0, \Diamond q) \;=\; \sup_{j\geq 0}\{ R(w_0,w_j) \wedge e(w_j,q) \} \;=\; \sup\{ \overbrace{0 \wedge 0}^{0}, \overbrace{1 \wedge (1/2)}^{1/2}, \overbrace{1 \wedge (2/3)}^{2/3}, \overbrace{1 \wedge (3/4)}^{3/4}, \dots \} \;=\; 1$$

$$e(w_j, \neg\neg p) = 1, \; \forall j \geq 1 \quad e(w_0, \Box\neg\neg p) = 1 \quad e(w_0, \neg\neg\Box p) = 0 \quad e(w_0, \; \Box\neg\neg p \to \neg\neg\Box p) = 0$$

Figure 1: A non-witnessed G-model $\mathfrak{M} = \langle W, R, e \rangle$.

$R : W \times W \to \{0,1\}$, we say that $\mathfrak{M}$ is *crisp*; in this case, $R \subseteq W \times W$. The map e is extended to arbitrary formulas as follows:

$$e(w, \bot) = 0 \qquad e(w, \alpha \star \beta) = e(w,\alpha) \star e(w,\beta), \; \star \in \{\wedge, \vee, \to\}$$
$$e(w, \Box\alpha) = \inf_{w' \in W}\{ R(w,w') \to e(w', \alpha) \} \quad e(w, \Diamond\alpha) = \sup_{w' \in W}\{ R(w,w') \wedge e(w', \alpha) \}.$$

Note that $e(w, \neg\alpha) = 1$ if $e(w,\alpha) = 0$ and $e(w, \neg\alpha) = 0$ if $e(w,\alpha) > 0$. A world w' is an R-successor of w iff $R(w,w') > 0$. If w has no R-successors, then $e(w, \Box\alpha) = 1$ and $e(w, \Diamond\alpha) = 0$ since, for every $w' \in W$, $R(w,w') \to e(w', \alpha) = 1$ and $R(w,w') \wedge e(w', \alpha) = 0$.

A G-model $\mathfrak{M} = \langle W, R, e \rangle$ is *witnessed*, and is called a GW-*model*, if the values of $e(w, \Box\alpha)$ and $e(w, \Diamond\alpha)$ are witnessed by some world w' in $\mathfrak{M}$, namely:

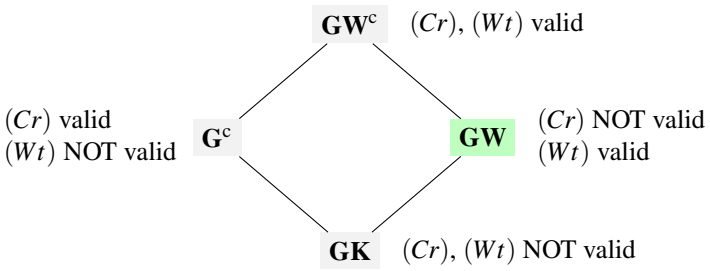
- $e(w, \Box\alpha) = r$ iff there is $w' \in W$ such that $R(w,w') \to e(w', \alpha) = r$;
- $e(w, \Diamond\alpha) = r$ iff there is $w' \in W$ such that $R(w,w') \wedge e(w', \alpha) = r$.

As a consequence, in the definitions of $e(w, \Box\alpha)$ and $e(w, \Diamond\alpha)$, the infimum and the supremum are just the minimum and the maximum. A non-witnessed model is shown in Fig. 1; whenever the value r of $R(w,w')$ or $e(w,p)$ is not displayed, it is understood that r is 0 (e.g., $R(w_i, w_j) = 0$, for every $i \geq 1$ and $j \geq 0$). It holds that $e(w_0, \Box p) = 0$, but the witnessing condition fails since there is no $w_j \in W$ such that $R(w_0, w_j) \to e(w_j, p) = 0$. Similarly $e(w_0, \Diamond q) = 1$, but there is no witness. When denoting a class of models, the superscript c means "crisp" (e.g., GW^{c}-model stands for "crisp GW-model").

A formula φ is *valid* in $\mathfrak{M} = \langle W, R, e \rangle$ iff $e(w, \varphi) = 1$ for every world w in W. If φ is not valid in $\mathfrak{M}$, we say that $\mathfrak{M}$ is a *countermodel* for φ; thus, $\mathfrak{M}$ contains a world w such that $e(w, \varphi) < 1$. Let $[0,1]_{\mathbf{Q}} = [0,1] \cap \mathbf{Q}$, where $\mathbf{Q}$ is the set of rational numbers; a model $\mathfrak{M} = \langle W, R, e \rangle$ is *discrete* if W is finite and the images of R and e are finite subsets of $[0,1]_{\mathbf{Q}}$. We stress that every finite G-model is witnessed.

Let **G** be the Gödel-Dummett Logic, obtained by extending Intuitionistic Propositional Logic **IPL** with the linearity axiom $(\alpha \to \beta) \vee (\beta \to \alpha)$. Gödel Modal Logic **GK** is obtained by adding to **G** the following axioms and rules ($\vdash$ refers to provability in **GK**):

$(K_\Box)$	$\Box(\alpha \to \beta) \to (\Box\alpha \to \Box\beta)$	$(K_\Diamond)$	$\Diamond(\alpha \vee \beta) \to (\Diamond\alpha \vee \Diamond\beta)$	$(F_\Diamond)$ $\neg\Diamond\bot$
(FS_1)	$\Diamond(\alpha \to \beta) \to (\Box\alpha \to \Diamond\beta)$	(FS_2)	$(\Diamond\alpha \to \Box\beta) \to \Box(\alpha \to \beta)$	
$(N_\Box)$	$\vdash \alpha$ implies $\vdash \Box\alpha$	$(N_\Diamond)$	$\vdash \alpha \to \beta$ implies $\vdash \Diamond\alpha \to \Diamond\beta$	



- (Cr) $\Box(\alpha \vee \beta) \to (\Box\alpha \vee \Diamond\beta)$
- (Wt) $\Box\neg\neg\alpha \to \neg\neg\Box\alpha$
- $\mathbf{G^c} = \mathbf{GK} + (Cr)$ (see [17])
- $\mathbf{GK} \subset \mathbf{G^c} \subset \mathbf{GW^c}$, $\mathbf{GK} \subset \mathbf{GW} \subset \mathbf{GW^c}$

Figure 2: Logics overview.

In [6] it is proved that **GK** is the set of formulas valid in every G-model. We introduce some logics extending **GK**, by providing a semantic characterization (see Fig. 2):

- $\mathbf{G^c}$ is the set of formulas valid in every $\mathrm{G^c}$-model (see [17]);
- **GW** is the set of formulas valid in every GW-model;
- $\mathbf{GW^c}$ is the set of formulas valid in every $\mathrm{GW^c}$-model (see [11]).

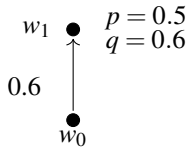
The logic $\mathbf{G^c}$ is obtained by adding the Crisp axiom $\Box(\alpha \vee \beta) \to (\Box\alpha \vee \Diamond\beta)$, which we call (Cr), to **GK** (see [17]). The logic $\mathbf{GW^c}$ was introduced in [11] and recently an axiomatization has been proposed in [18]. By definition, $\mathbf{GK} \subseteq \mathbf{G^c} \subseteq \mathbf{GW^c}$ and $\mathbf{GK} \subseteq \mathbf{GW} \subseteq \mathbf{GW^c}$; we show that all the inclusions are strict. Let us consider the instance $\varphi = \Box(p \vee q) \to (\Box p \vee \Diamond q)$ of (Cr) and let $\mathfrak{M} = \langle W, R, e\rangle$ be the GW-model in Fig. 3. Since $e(w_0, \varphi) = 0.6$, $\mathfrak{M}$ is a countermodel for φ, and this certifies that $\varphi \notin \mathbf{GW}$; accordingly, axiom (Cr) is not valid in **GW**. Let (Wt) be the formula schema $\Box\neg\neg\alpha \to \neg\neg\Box\alpha$; we show that (Wt) is valid in **GW**. Let $\mathfrak{M} = \langle W, R, e\rangle$ be a GW-model and $w \in W$. If $e(w, \neg\neg\Box\alpha) = 1$, we immediately get $e(w, (Wt)) = 1$. Otherwise, $e(w, \neg\neg\Box\alpha) = 0$, hence $e(w, \Box\alpha) = 0$. Since $\mathfrak{M}$ is witnessed, there exists $w' \in W$ such that $R(w, w') \to e(w', \alpha) = 0$; it follows that $e(w', \alpha) = 0$, hence $e(w', \neg\neg\alpha) = 0$. This implies $e(w, \Box\neg\neg\alpha) = 0$, hence $e(w, (Wt)) = 1$. This proves that (Wt) is valid in **GW**. Since every finite model is witnessed, a countermodel for (Wt) must be infinite. The instance $\Box\neg\neg p \to \neg\neg\Box p$ of (Wt) is not valid in the $\mathrm{G^c}$-model $\mathfrak{M}$ in Fig. 1 (see, e.g., [11]), thus (Wt) is not valid in $\mathbf{G^c}$; this also ascertains that neither $\mathbf{G^c}$ nor **GK** enjoy the finite model property.

We show that the (classical) modal logic **K** can be embedded into each of the logics in Fig. 2, by exploiting the translation $*$ introduced in [15]. We stress that a model for the logic **K** (a K-*model*) can be viewed as a $\mathrm{GW^c}$-model whose evaluation function takes values only in the set $\{0, 1\}$; accordingly, $\mathbf{GW^c} \subseteq \mathbf{K}$. We assume that $\Diamond$ is not a primitive connective in the language of **K** (actually, in **K** the formula $\Diamond\varphi$ can be defined as $\neg\Box\neg\varphi$). Let α be a formula of **K**; the formula α^* is obtained by replacing each propositional variable p occurring in α with $\neg\neg p$. As shown in the following lemma, the evaluation of a formula α^* at any world of a G-model is binary (0 or 1); this is essentially due to the fact that propositional variables in α^* are double negated and $\Diamond$ does not occur in α^*.

Lemma 2.1. *Let α be a formula of* **K**.

(i) If $\alpha \in \mathbf{K}$ then $\alpha^* \in \mathbf{GK}$.

(ii) If $\alpha^* \in \mathbf{GW^c}$ then $\alpha \in \mathbf{K}$.



$W = \{w_0,\ w_1\}$
$R(w_0,w_0) = R(w_1,w_1) = 0,\ R(w_0,w_1) = 0.6$
$e(w_0,p) = e(w_0,q) = 0$
$e(w_1,p) = 0.5,\ e(w_1,q) = 0.6$

$$
\begin{aligned}
e(w_0,\Box(p\vee q)) &= \inf\{R(w_0,w_0)\to e(w_0,p\vee q), R(w_0,w_1)\to e(w_1,p\vee q)\} = \inf\{0\to 0, 0.6\to 0.6\} = 1\\
e(w_0,\Box p) &= \inf\{R(w_0,w_0)\to e(w_0,p), R(w_0,w_1)\to e(w_1,p)\} = \inf\{0\to 0, 0.6\to 0.5\} = 0.5\\
e(w_0,\Diamond q) &= \sup\{R(w_0,w_0)\wedge e(w_0,q), R(w_0,w_1)\wedge e(w_1,q)\} = \sup\{0\wedge 0, 0.6\wedge 0.6\} = 0.6\\
e(w_0,\varphi) &= e(w_0,\Box(p\vee q))\to e(w_0,\Box p\vee\Diamond q) = 1\to 0.6 = 0.6
\end{aligned}
$$

Figure 3: A countermodel for the formula $\varphi = \Box(p\vee q)\to(\Box p\vee\Diamond q)$ (instance of (Cr)).

Proof. The proof of point (ii) is immediate, since $\mathbf{GW}^{\mathrm{c}}\subseteq\mathbf{K}$ and the formula $\alpha\leftrightarrow\alpha^*$ is valid in $\mathbf{K}$. We prove point (i). Let us assume that $\alpha^*\notin\mathbf{GK}$. There exists a G-model $\mathfrak{M} = \langle W,R,e\rangle$ and a world w in W such that $e(w,\alpha^*) < 1$. Let $\mathfrak{M}_1 = \langle W,R_1,e_1\rangle$ be the K-model defined as follows:

$$
R_1(w,w') = \begin{cases} 1 & \text{if } R(w,w') > 0\\ 0 & \text{otherwise}\end{cases}
\qquad\qquad
e_1(w,p) = \begin{cases} 1 & \text{if } e(w,p) > 0\\ 0 & \text{otherwise}\end{cases}
$$

Let $w\in W$ and let φ be any formula of $\mathbf{K}$; by induction on the size of φ, we prove that:

(A) $e_1(w,\varphi) = e(w,\varphi^*)$.

The case $\varphi\in\mathscr{V}\cup\{\bot\}$ is immediate. Let $\varphi = \Box\psi$; note that $\varphi^* = \Box\psi^*$. Assume $e_1(w,\Box\psi) = 0$. There exists $w'\in W$ such that $R_1(w,w') = 1$ and $e_1(w',\psi) = 0$. Then, $R(w,w') > 0$ and, by the induction hypothesis, $e(w',\psi^*) = 0$. It follows that $R(w,w')\to e(w',\psi^*) = 0$, hence $e(w,\Box\psi^*) = 0$. Let $e_1(w,\Box\psi) = 1$ and assume, by contradiction, $e(w,\Box\psi^*) < 1$. By the induction hypothesis, for every $w'\in W$ it holds that $e(w',\psi^*)\in\{0,1\}$, hence $R(w,w')\to e(w',\psi^*)\in\{0,1\}$ and $e(w,\Box\psi^*)\in\{0,1\}$. Thus, $e(w,\Box\psi^*) = 0$ and there is $w'\in W$ such that $R(w,w')\to e(w',\psi^*) = 0$. We get $R(w,w') > 0$ and $e(w',\psi^*) = 0$, hence $R_1(w,w') = 1$ and $e_1(w',\psi) = 0$, contradicting the assumption $e_1(w,\Box\psi) = 1$. Since the assumption $e(w,\Box\psi^*) < 1$ yields a contradiction, we conclude $e(w,\Box\psi^*) = 1$. The proof of (A) in the remaining cases easily follows from the induction hypothesis. Since $e(w,\alpha^*) < 1$, from (A) it follows that $e_1(w,\alpha) = 0$. Accordingly, $\alpha\notin\mathbf{K}$, and this proves (i). □

As a consequence of Lemma 2.1, we get:

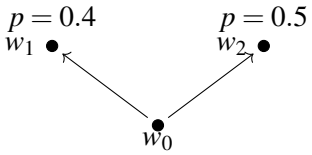
Proposition 2.2. *Let α be a formula of* **K** *and let L be any of the logics* **GK**, $\mathbf{G}^{\mathrm{c}}$, **GW** *and* $\mathbf{GW}^{\mathrm{c}}$. *Then,* $\alpha\in\mathbf{K}$ *iff* $\alpha^*\in L$.

We conclude this section by discussing the non-interdefinability of $\Box$ and $\Diamond$. This was established for $\mathbf{G}^{\mathrm{c}}$ in [17] using algebraic semantics. Here, we instead employ GW^{c}-models to extend the result to any modal logic contained in $\mathbf{GW}^{\mathrm{c}}$. We prove that:

(P1) there is no $\Box$-free formula φ such that $\Box p\leftrightarrow\varphi$ is valid in $\mathbf{GW}^{\mathrm{c}}$;

(P2) there is no $\Diamond$-free formula ψ such that $\Diamond p\leftrightarrow\psi$ is valid in $\mathbf{GW}^{\mathrm{c}}$.

Lemma 2.3. *Let $\mathfrak{M}^* = \langle W,R,e\rangle$ be the* GW^{c}*-model defined in Fig. 4.*



$W = \{w_0, w_1, w_2\}$
$R = \{(w_0, w_1), (w_0, w_2)\}$
$e(w_1, p) = 0.4, e(w_2, p) = 0.5$

Figure 4: The GW^c-model $\mathfrak{M}^* = \langle W, R, e \rangle$.

(i) For every $\Box$-free formula φ, $e(w_0, \varphi) \in \{0, 0.5, 1\}$.

(ii) For every $\Diamond$-free formula ψ, $e(w_0, \psi) \in \{0, 0.4, 1\}$.

Proof. One can easily prove that, for every formula α, the following facts hold:

(1) $e(w_1, \alpha) \in \{0, 0.4, 1\}$.

(2) $e(w_2, \alpha) = \Phi(e(w_1, \alpha))$, where $\Phi(0) = 0$, $\Phi(0.4) = 0.5$, $\Phi(1) = 1$.

We prove (i), by induction on the size of φ. If $\varphi \in \mathcal{V} \cup \{\bot\}$, we have $e(w_0, \varphi) = 0$. The cases $\varphi = \alpha \wedge \beta$, $\varphi = \alpha \vee \beta$ and $\varphi = \alpha \to \beta$ easily follow from the induction hypothesis. Let $\varphi = \Diamond\alpha$. By (1) and (2), one of the following properties (a)–(c) holds:

(a) $e(w_1, \alpha) = e(w_2, \alpha) = 0$.

(b) $e(w_1, \alpha) = 0.4$ and $e(w_2, \alpha) = 0.5$.

(c) $e(w_1, \alpha) = e(w_2, \alpha) = 1$.

In case (a) we get $e(w_0, \Diamond\alpha) = 0$, in case (b) $e(w_0, \Diamond\alpha) = 0.5$, in case (c) $e(w_0, \Diamond\alpha) = 1$; this concludes the proof of (i). The proof of (ii) is similar. □

Proposition 2.4. *Properties (P1) and (P2) hold.*

Proof. Let us assume, by contradiction, that property (P1) does not hold. Then, there exists a $\Box$-free formula φ such that $\Box p \leftrightarrow \varphi$ is valid in **GW**^{c}. As a consequence, in the GW^c-model $\mathfrak{M}^*$ in Fig. 4 we have $e(w_0, \Box p \leftrightarrow \varphi) = 1$. Since $e(w_0, \Box p) = 0.4$, we get $e(w_0, \varphi) = 0.4$, in contradiction with Lemma 2.3(i). This proves that (P1) holds. The proof of (P2) is similar. □

Proposition 2.5. *$\Box$ and $\Diamond$ are not interdefinable in any modal logic L contained in* **GW**^{c}.

Proof. Let $L \subseteq$ **GW**^{c} and assume that there is a $\Box$-free formula φ such that $\Box p \leftrightarrow \varphi \in L$. Then, $\Box p \leftrightarrow \varphi \in$ **GW**^{c}, in contradiction with Lemma 2.4. The proof for $\Diamond$ is analogous. □

3 The Calculus $\mathscr{C}_{\mathrm{GW}}$

Following [3, 11], we introduce the refutation calculus $\mathscr{C}_{\mathrm{GW}}$ for **GW**, namely, a calculus to certify that a formula is not valid in any GW-model. The calculus exploits the constraint language $\mathscr{L}_c$ defined over a

$$\lhd \in \{<, \leq\}, \quad \rhd \in \{>, \geq\} \qquad \frac{}{\Gamma}\,\mathrm{Ax} \quad \text{if } \mathrm{At}^+(\Gamma) \text{ is not consistent}$$

$$\frac{w:\alpha \rhd t, w:\beta \rhd t, \Gamma}{w:\alpha \wedge \beta \rhd t, \Gamma}\wedge\rhd \qquad \frac{w:\alpha \lhd t, \Gamma \qquad w:\beta \lhd t, \Gamma}{w:\alpha \wedge \beta \lhd t, \Gamma}\wedge\lhd$$

$$\frac{w:\alpha \lhd t, w:\beta \lhd t, \Gamma}{w:\alpha \vee \beta \lhd t, \Gamma}\vee\lhd \qquad \frac{w:\alpha \rhd t, \Gamma \qquad w:\beta \rhd t, \Gamma}{w:\alpha \vee \beta \rhd t, \Gamma}\vee\rhd$$

$$\frac{w:\alpha > w:p, w:p < t, \Gamma}{w:\alpha \to p < t, \Gamma}\to_{\mathrm{At}}< \ (\dagger) \qquad \frac{w:\alpha > w:q, w:\beta \leq w:q, w:q < t, \Gamma}{w:\alpha \to \beta < t, \Gamma}\to< \ (\dagger)$$

$$\frac{t \geq 1, \Gamma \qquad w:\alpha > w:p, w:p \leq t, \Gamma}{w:\alpha \to p \leq t, \Gamma}\to_{\mathrm{At}}\leq \ (\dagger)$$

$$\frac{t \geq 1, \Gamma \qquad w:\alpha > w:q, w:\beta \leq w:q, w:q \leq t, \Gamma}{w:\alpha \to \beta \leq t, \Gamma}\to\leq \ (\dagger)$$

$$\frac{w:\alpha \leq w:p, 1 \rhd t, \Gamma \qquad w:p \rhd t, \Gamma}{w:\alpha \to p \rhd t, \Gamma}\to_{\mathrm{At}}\rhd (\dagger)$$

$$\frac{w:\alpha \ \leq w:q, w:\beta \geq w:q, 1 \rhd t, \Gamma \qquad w:\beta \rhd t, \Gamma}{w:\alpha \to \beta \rhd t, \Gamma}\to\rhd (\dagger)$$

$$\mathrm{At}^+(\Gamma) \ = \ \mathrm{At}(\Gamma) \cup \{1 > t \mid w:\Box\alpha > t \in \Gamma\} \cup \{0 < t \mid w:\Diamond\alpha < t \in \Gamma\}$$

$(\dagger)$ $p \in \mathscr{V} \cup \{\bot\}$, $\beta \notin \mathscr{V} \cup \{\bot\}$, q is a new propositional variable

Figure 5: The calculus $\mathscr{C}_{\mathrm{GW}}$, propositional rules ($\Gamma$ is a multiset of constraints).

countable set of labels, used to represent the worlds of a GW-model. In the definitions below w, w' are labels, φ is a formula, $p \in \mathscr{V}$, R is a designated relation symbol, representing the accessibility relation.

$$\begin{array}{rcl} \text{labelled formula} & := & w:\varphi \\ \text{atomic c-term } t & := & 0 \mid 1 \mid w:p \mid w:\bot \mid R(w,w') \\ \text{c-term } u & := & t \mid w:\varphi \mid R(w,w') \wedge w':\varphi \mid R(w,w') \to w':\varphi \\ \text{constraint } \chi & := & u \blacktriangledown t \qquad \blacktriangledown \in \{<, \leq, >, \geq\} \end{array}$$

If $w:\varphi$ occurs in a c-term u, we say that u has label w. A constraint of the form $t \blacktriangledown t$ is called *atomic*; if $\chi = w:\varphi \blacktriangledown t$ is non-atomic and $\sharp$ is the main connective of φ, we say that χ is a $\sharp$-*constraint*. Constraints where u is $R(w,w') \wedge w':\varphi$ or $R(w,w') \to w':\varphi$ are called *R-constraints*. Given a multiset of constraints Γ, by $\mathrm{At}(\Gamma)$ we denote the set of atomic constraints in Γ. Let $\mathfrak{M} = \langle W, R, e\rangle$ be a GW-model. An $\mathfrak{M}$-*interpretation* of $\mathscr{L}_c$ is a function $\mathscr{I}$ mapping labels of $\mathscr{L}_c$ to W. We extend $\mathscr{I}$ to c-terms as follows:

$$\mathscr{I}(k) = k, \ k \in \{0,1\} \qquad \mathscr{I}(w:\varphi) = e(\mathscr{I}(w), \varphi) \qquad \mathscr{I}(R(w,w')) = R(\mathscr{I}(w), \mathscr{I}(w'))$$

$$\mathscr{I}(R(w,w') \star w':\varphi) = R(\mathscr{I}(w), \mathscr{I}(w')) \star e(\mathscr{I}(w'), \varphi), \ \star \in \{\wedge, \to\}$$

Note that, for every c-term u, $\mathscr{I}(u)$ belongs to $[0,1]$. We introduce the relations $\models_{\mathscr{I}}$ and $\models$, where $\mathfrak{M}$ is a GW-model, $\mathscr{I}$ an $\mathfrak{M}$-interpretation, Γ a multiset of constraints.

$$\frac{R(w,w_1) \to w_1 : \alpha \lhd t,\ \Phi^{\Box,\Diamond}(\Gamma,w,w_1),\ \Gamma}{w : \Box\alpha \lhd t, \Gamma}\ \Box\lhd \qquad \frac{R(w,w_1) \wedge w_1 : \alpha \rhd t,\ \Phi^{\Box,\Diamond}(\Gamma,w,w_1),\ \Gamma}{w : \Diamond\alpha \rhd t, \Gamma}\ \Diamond\rhd$$

$$\frac{R(w,w') \rhd t,\ w' : \alpha \rhd t, \Gamma}{R(w,w') \wedge w' : \alpha \rhd t, \Gamma}\ R\wedge\rhd \qquad \frac{R(w,w') \lhd t, \Gamma \qquad w' : \alpha \lhd t, \Gamma}{R(w,w') \wedge w' : \alpha \lhd t, \Gamma}\ R\wedge\lhd$$

$$\frac{w' : \alpha < R(w,w'),\ w' : \alpha < t, \Gamma}{R(w,w') \to w' : \alpha < t, \Gamma}\ R\to< \qquad \frac{t \geq 1, \Gamma \qquad w' : \alpha < R(w,w'),\ w' : \alpha \leq t, \Gamma}{R(w,w') \to w' : \alpha \leq t, \Gamma}\ R\to\leq$$

$$\frac{w' : \alpha \geq R(w,w'),\ 1 \rhd t, \Gamma \qquad w' : \alpha \rhd t, \Gamma}{R(w,w') \to w' : \alpha \rhd t, \Gamma}\ R\to\rhd$$

$\Phi^{\Box,\Diamond}(\Gamma,w,w_1) = \{R(w,w_1) \to w_1 : \beta \rhd' t \mid w : \Box\beta \rhd' t \in \Gamma\} \cup \{R(w,w_1) \wedge w_1 : \beta \lhd' t \mid w : \Diamond\beta \lhd' t \in \Gamma\}$
w_1 is a new label, $\lhd, \lhd' \in \{<, \leq\}$ and $\rhd, \rhd' \in \{>, \geq\}$.

Figure 6: The calculus $\mathscr{C}_{\mathrm{GW}}$: modal rules and R-rules (Γ is a multiset of constraints).

- $\mathfrak{M} \models_{\mathscr{I}} u \blacktriangledown t$ iff $\mathscr{I}(u) \blacktriangledown \mathscr{I}(t)$;
- $\mathfrak{M} \models_{\mathscr{I}} \Gamma$ iff $\mathfrak{M} \models_{\mathscr{I}} \chi$, for every $\chi \in \Gamma$;
- $\mathfrak{M} \models \Gamma$ iff $\mathfrak{M} \models_{\mathscr{I}} \Gamma$ for some $\mathfrak{M}$-interpretation $\mathscr{I}$.

A *substitution* σ is a function mapping each atomic c-term of the form $w : p$ or $R(w,w')$ to a rational number in $[0,1]_{\mathbf{Q}}$; σ is extended to all the atomic c-terms by setting $\sigma(k) = k$, for $k \in \{0,1\}$, and $\sigma(w : \bot) = 0$. Let Γ_{at} be a set of atomic constraints. By $\sigma(\Gamma_{\mathrm{at}})$ we denote the set of constraints obtained by replacing every atomic c-term t occurring in Γ_{at} with $\sigma(t)$. Note that $\sigma(\Gamma_{\mathrm{at}})$ is a set of rational constraints of the form $r_1 \blacktriangledown r_2$, with r_1 and r_2 in $[0,1]_{\mathbf{Q}}$; if all the constraints in $\sigma(\Gamma_{\mathrm{at}})$ hold, σ is a *solution* to Γ_{at}. The set Γ_{at} is *consistent* iff it admits at least one solution. We remark that consistency of Γ_{at} can be checked by a Constraint Solver over Q: one has to abstract the c-terms $w : p$ and $R(w,w')$ occurring in Γ_{at} by introducing new variables ranging over $[0,1]_{\mathbf{Q}}$, and then check the consistency of the obtained constraints by exploiting the solver.

Hereafter, in constraints $\lhd$ and $\lhd'$ are metasymbols ranging over $\{<, \leq\}$; similarly $\rhd, \rhd' \in \{>, \geq\}$. The rules of the calculus $\mathscr{C}_{\mathrm{GW}}$ are displayed in Fig. 5 (propositional rules) and Fig. 6 (modal rules and R-rules). The *main constraint* of a rule application is the constraint displayed in the conclusion. Rules for implication, having a main constraint of the kind $w : \alpha \to \beta \blacktriangledown t$, are defined according to the structure of β. Let us consider the rule $\to_{\mathrm{At}}<$, having main constraint $w : \alpha \to p < t$ ($p \in \mathscr{V} \cup \{\bot\}$). The premise contains the constraints $w : \alpha > w : p$ and $w : p < t$. This reflects the fact that, given a GW-model $\mathfrak{M}$, a world w in $\mathfrak{M}$ and an $\mathfrak{M}$-interpretation $\mathscr{I}$, if $e(\mathscr{I}(w), \alpha \to p) < \mathscr{I}(t)$, it holds that $e(\mathscr{I}(w), \alpha) > e(\mathscr{I}(w), p)$, $e(\mathscr{I}(w), \alpha \to p) = e(\mathscr{I}(w), p)$ and $e(\mathscr{I}(w), p) < \mathscr{I}(t)$. This rule cannot be generalized to any $w : \alpha \to \beta < t$ since the constraint $w : \alpha > w : \beta$ is not allowed if $\beta \notin \mathscr{V} \cup \{\bot\}$. This case is covered by rule $\to<$, which introduces a new propositional variable q behaving in w as β, and $w : \alpha > w : \beta$ is replaced by $w : \alpha > w : q$. The correspondence between β and q is set by the constraint $w : \beta \leq w : q$, while the converse constraint $w : \beta \geq w : q$ can be omitted. There are two modal rules (see Fig. 6), the former having main constraint $w : \Box\alpha \lhd t$ and the latter $w : \Diamond\alpha \rhd t$; both rules introduce a new label w_1. R-rules (see Fig. 6) are similar to the corresponding propositional rules. The definitions of *tree* and *derivation* of $\mathscr{C}_{\mathrm{GW}}$ are the usual ones (see, e.g., [19]). We remark that $\mathscr{C}_{\mathrm{GW}}$ has

$$c = w_0 : q_0 \quad \Delta_0 = \{ w_0 : \Box p > c,\ c \geq 1,\ c < 1 \} \quad \Delta_1 = \{ w_0 : \bot \leq c,\ c < 1 \}$$
$$\Delta_2 = \Delta_1 \cup \{ w_0 : \Box p > c,\ R(w_0, w_1) > w_0 : \bot \} \qquad \Delta_3 = \Delta_2 \cup \{\ w_1 : p \leq w_1 : \bot,\ 1 > w_0 : \bot \}$$
$$\Delta_4 = \Delta_2 \cup \{ R(w_0, w_1) \to w_1 : p > c,\ w_1 : \bot > w_0 : \bot \}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{w_1 : p \geq R(w_0,w_1),\ 1 > c,\ \Delta_3}\ \text{Ax} \quad \dfrac{}{w_1 : p > c,\ \Delta_3}\ \text{Ax}}{R(w_0,w_1) \to w_1 : p > c,\ \Delta_3}\ R \to \rhd \quad \dfrac{}{\Delta_4}\ \text{Ax}}{R(w_0,w_1) \to w_1 : p > c,\ w_1 : \neg p > w_0 : \bot,\ \Delta_2}\ \to_{\text{At}} \rhd}{w_0 : \Box p > c,\ R(w_0,w_1) \to w_1 : p > c,\ R(w_0,w_1) \wedge w_1 : \neg p > w_0 : \bot,\ \Delta_1}\ R \wedge \rhd}{w_0 : \Box p > c,\ w_0 : \Diamond \neg p > w_0 : \bot,\ \Delta_1}\ \Diamond \rhd$$
$$\dfrac{\dfrac{\dfrac{}{\Delta_0}\ \text{Ax} \qquad \text{(above)}}{w_0 : \Box p > c,\ w_0 : \neg \Diamond \neg p \leq c,\ c < 1}\ \to_{\text{At}} \leq}{w_0 : \Box p \to \neg \Diamond \neg p < 1}\ \to <$$

Figure 7: A derivation of $w_0 : \Box p \to \neg \Diamond \neg p < 1$.

the *subformula property*, namely: if $\mathscr{T}$ is a tree of $\mathscr{C}_{\text{GW}}$ having Γ_0 as root, every formula occurring in $\mathscr{T}$ inside a constraint (e.g., the formula φ in the constraint $R(w,w') \to w' : \varphi$) is a subformula of a formula in Γ_0 or a new propositional variable. By $\vdash_{\text{GW}} \Gamma$ we mean that there exists a derivation of Γ. In the rest of this section we show that $\mathscr{C}_{\text{GW}}$ is a sound and complete refutation calculus for **GW**. Formally, soundness and completeness are defined as follows, where Γ is a multiset of constraints of the form $w : \varphi \blacktriangledown t$:

- (Soundness) if $\vdash_{\text{GW}} \Gamma$, then $\mathfrak{M} \not\models \Gamma$, for every GW-model $\mathfrak{M}$;
- (Completeness) if $\mathfrak{M} \not\models \Gamma$, for every GW-model $\mathfrak{M}$, then $\vdash_{\text{GW}} \Gamma$.

We stress that, if rule Ax checked the consistency of $\text{At}(\Gamma)$ instead of $\text{At}^+(\Gamma)$, the calculus $\mathscr{C}_{\text{GW}}$ would not be complete. Indeed, let $\Gamma = \{w : \Box p > w : p,\ w : p \leq 1,\ w : p \geq 1\}$. Clearly, there is no GW-model $\mathfrak{M}$ such that $\mathfrak{M} \models \Gamma$ hence, by completeness, Γ must be provable in $\mathscr{C}_{\text{GW}}$. To build a derivation of Γ, one can only exploit rule Ax. Note that $\text{At}(\Gamma) = \{w : p \leq 1,\ w : p \geq 1\}$ is consistent (any substitution σ such that $\sigma(w : p) = 1$ is a solution); thus, if Ax checked the consistency of $\text{At}(\Gamma)$, Γ could not be proved. Instead, Γ is proved since Ax evaluates the set $\text{At}^+(\Gamma) = \text{At}(\Gamma) \cup \{1 > w : p\}$, which is not consistent.

An example of derivation is shown in Fig. 7; the main constraint of a rule application is underlined.

Let $\mathfrak{M} = \langle W, R, e \rangle$ be a GW-model and $\mathscr{I}$ an $\mathfrak{M}$-interpretation. Let $p_0 \in \mathscr{V}$ and α a formula not containing p_0, let $w_0 \in \mathscr{L}_c$ and $w^\star \in W$. The evaluation function $e[p_0 := \alpha]$ and the $\mathfrak{M}$-interpretation $\mathscr{I}[w_0 := w^\star]$ are defined as follows ($w \in W$ and $p \in \mathscr{V}$):

$$e[p_0 := \alpha](w,p) = \begin{cases} e(w,\alpha) & \text{if } p = p_0 \\ e(w,p) & \text{otherwise} \end{cases} \qquad \mathscr{I}[w_0 := w^\star](w) = \begin{cases} w^* & \text{if } w = w_0 \\ \mathscr{I}(w) & \text{otherwise} \end{cases}$$

Let $e' = e[p_0 := \alpha]$ and $w \in W$; one can prove that, for every formula φ not containing p_0, $e'(w,\varphi) = e(w,\varphi)$. Accordingly, $e'(w, p_0 \leftrightarrow \alpha) = 1$, and this implies that $\langle W, R, e' \rangle$ is a GW-model. Let $\mathscr{I}' = \mathscr{I}[w_0 := w^\star]$ and let χ be a constraint not containing w_0; one can prove that $\mathfrak{M} \models_{\mathscr{I}'} \chi$ iff $\mathfrak{M} \models_{\mathscr{I}} \chi$.

Lemma 3.1. *Let ρ be an instance of a rule of $\mathscr{C}_{\text{GW}}$ and let Γ be the conclusion of ρ. If there is a* GW*-model $\mathfrak{M}$ such that $\mathfrak{M} \models \Gamma$, then there exist a* GW*-model $\mathfrak{M}'$ and a premise Γ' of ρ such that $\mathfrak{M}' \models \Gamma'$.*

Proof. Let ρ be an instance of the rule Ax; since the rule has no premises, we have to show that there is no $\mathfrak{M}$ such that $\mathfrak{M} \models \Gamma$. Let us assume, by contradiction, that there is a GW-model $\mathfrak{M} = \langle W, R, e \rangle$

and an $\mathfrak{M}$-interpretation $\mathscr{I}$ such that $\mathfrak{M} \models_{\mathscr{I}} \Gamma$. If $w : \Box\alpha > t \in \Gamma$, then $e(\mathscr{I}(w), \Box\alpha) > \mathscr{I}(t)$, hence $1 > \mathscr{I}(t)$, which implies $\mathfrak{M} \models_{\mathscr{I}} 1 > t$. Similarly, if $w : \Diamond\alpha < t \in \Gamma$, then $\mathfrak{M} \models_{\mathscr{I}} 0 < t$. This proves that $\mathfrak{M} \models_{\mathscr{I}} \mathrm{At}^{+}(\Gamma)$. By exploiting R and e, one can define a solution to $\mathrm{At}^{+}(\Gamma)$; accordingly, $\mathrm{At}^{+}(\Gamma)$ is consistent, a contradiction. We conclude that $\mathfrak{M}$ does not exist.

Let $\rho \neq \mathrm{Ax}$ and let us assume $\mathfrak{M} \models_{\mathscr{I}} \Gamma$, where $\mathfrak{M} = \langle W, R, e\rangle$ is a GW-model and $\mathscr{I}$ is an $\mathfrak{M}$-interpretation; we prove that there exist a GW-model $\mathfrak{M}'$ and a premise Γ' of ρ such that $\mathfrak{M}' \models \Gamma'$. We only detail some representative cases, by showing the corresponding rule application.

$$\dfrac{\overbrace{w : \alpha > w : q, w : \beta \leq w : q, w : q < t, \Delta}^{\Gamma_1}}{\underbrace{w : \alpha \to \beta < t, \Delta}_{\Gamma}} \to< \qquad \begin{array}{l} \beta \notin \mathscr{V} \cup \{\bot\} \\ q \text{ does not occur in } \Gamma \end{array}$$

By the hypothesis, $\mathfrak{M} \models_{\mathscr{I}} w : \alpha \to \beta < t$ and $\mathfrak{M} \models_{\mathscr{I}} \Delta$. This implies that $e(\mathscr{I}(w), \alpha \to \beta) < \mathscr{I}(t)$, hence $e(\mathscr{I}(w), \alpha) > e(\mathscr{I}(w), \beta)$ and $e(\mathscr{I}(w), \alpha \to \beta) = e(\mathscr{I}(w), \beta)$, thus $e(\mathscr{I}(w), \beta) < \mathscr{I}(t)$. Let $\mathfrak{M}' = \langle W, R, e[q := \beta]\rangle$; we remark that $\mathfrak{M}'$ is a GW-model. Since $\mathfrak{M} \models_{\mathscr{I}} \Delta$ and q does not occur in Δ, we get $\mathfrak{M}' \models_{\mathscr{I}} \Delta$. It is easy to check that $\mathfrak{M}' \models_{\mathscr{I}} w : \beta \leq w : q$ and $\mathfrak{M}' \models_{\mathscr{I}} w : \alpha > w : q$ and $\mathfrak{M}' \models_{\mathscr{I}} w : q < t$; we conclude $\mathfrak{M}' \models_{\mathscr{I}} \Gamma_1$.

$$\dfrac{\overbrace{t \geq 1, \Delta}^{\Gamma_1} \qquad \overbrace{w : \alpha > w : q, w : \beta \leq w : q, w : q \leq t, \Delta}^{\Gamma_2}}{\underbrace{w : \alpha \to \beta \leq t, \Delta}_{\Gamma}} \to\leq \qquad \begin{array}{l} \beta \notin \mathscr{V} \cup \{\bot\} \\ q \text{ does not occur in } \Gamma \end{array}$$

By the hypothesis, $\mathfrak{M} \models_{\mathscr{I}} w : \alpha \to \beta \leq t$ and $\mathfrak{M} \models_{\mathscr{I}} \Delta$, hence $e(\mathscr{I}(w), \alpha \to \beta) \leq \mathscr{I}(t)$. Let us assume $e(\mathscr{I}(w), \alpha) \leq e(\mathscr{I}(w), \beta)$; then, $e(\mathscr{I}(w), \alpha \to \beta) = 1$, which implies $1 \leq \mathscr{I}(t)$, thus $\mathscr{I}(t) = 1$. We get $\mathfrak{M} \models_{\mathscr{I}} t \geq 1$, hence $\mathfrak{M} \models_{\mathscr{I}} \Gamma_1$. Assume $e(\mathscr{I}(w), \alpha) > e(\mathscr{I}(w), \beta)$; then, $e(\mathscr{I}(w), \alpha \to \beta) = e(\mathscr{I}(w), \beta)$, hence $\mathfrak{M} \models_{\mathscr{I}} w : \beta \leq t$. Let $\mathfrak{M}' = \langle W, R, e[q := \beta]\rangle$; note that $\mathfrak{M}'$ is a GW-model. Since $\mathfrak{M} \models_{\mathscr{I}} \Delta$ and q does not occur in Δ, we get $\mathfrak{M}' \models_{\mathscr{I}} \Delta$. Moreover, one can easily check that $\mathfrak{M}' \models_{\mathscr{I}} w : \beta \leq w : q$ and $\mathfrak{M}' \models_{\mathscr{I}} w : \alpha > w : q$ and $\mathfrak{M}' \models_{\mathscr{I}} w : q \leq t$; we conclude $\mathfrak{M}' \models_{\mathscr{I}} \Gamma_2$.

$$\dfrac{\overbrace{w : \alpha \leq w : q, w : \beta \geq w : q, 1 \rhd t, \Delta}^{\Gamma_1} \qquad \overbrace{w : \beta \rhd t, \Delta}^{\Gamma_2}}{\underbrace{w : \alpha \to \beta \rhd t, \Delta}_{\Gamma}} \to \rhd \qquad \begin{array}{l} \beta \notin \mathscr{V} \cup \{\bot\} \\ q \text{ does not occur in } \Gamma \end{array}$$

By the hypothesis, $\mathfrak{M} \models_{\mathscr{I}} w : \alpha \to \beta \rhd t$ and $\mathfrak{M} \models_{\mathscr{I}} \Delta$, hence $e(\mathscr{I}(w), \alpha \to \beta) \rhd \mathscr{I}(t)$. Let us assume $e(\mathscr{I}(w), \alpha) > e(\mathscr{I}(w), \beta)$; then $e(\mathscr{I}(w), \alpha \to \beta) = e(\mathscr{I}(w), \beta)$, hence $\mathfrak{M} \models_{\mathscr{I}} \Gamma_2$. Let us assume $e(\mathscr{I}(w), \alpha) \leq e(\mathscr{I}(w), \beta)$; then, $e(\mathscr{I}(w), \alpha \to \beta) = 1$, which implies $1 \rhd \mathscr{I}(t)$, thus $\mathfrak{M} \models_{\mathscr{I}} 1 \rhd t$. Let $\mathfrak{M}' = \langle W, R, e[q := \alpha]\rangle$; note that $\mathfrak{M}'$ is a GW-model. Reasoning as in the previous cases, we can prove that $\mathfrak{M}' \models_{\mathscr{I}} \Gamma_1$.

$$\dfrac{\overbrace{R(w, w_1) \to w_1 : \alpha \lhd t, \Phi^{\Box,\Diamond}(\Gamma, w, w_1), \Delta}^{\Gamma_1}}{\underbrace{w : \Box\alpha \lhd t, \Delta}_{\Gamma}} \Box\lhd \qquad w_1 \text{ does not occur in } \Gamma$$

By the hypothesis, $\mathfrak{M} \models_{\mathscr{I}} w : \Box\alpha \lhd t$ and $\mathfrak{M} \models_{\mathscr{I}} \Delta$, hence $e(\mathscr{I}(w), \Box\alpha) \lhd \mathscr{I}(t)$. Since $\mathfrak{M}$ is a GW-model, there exists a world $w^\star$ of $\mathfrak{M}$ such that $R(\mathscr{I}(w), w^\star) \to e(w^\star, \alpha) = e(\mathscr{I}(w), \Box\alpha)$. Let $\mathscr{I}' = \mathscr{I}[w_1 := w^\star]$. We prove that:

(i) $\mathfrak{M} \models_{\mathscr{I}'} \Delta$.

(ii) $\mathfrak{M} \models_{\mathscr{I}'} R(w,w_1) \to w_1 : \alpha \lhd t$.

(iii) $\mathfrak{M} \models_{\mathscr{I}'} \Phi^{\Box,\Diamond}(\Gamma, w, w_1)$.

Point (i) follows from the fact that $\mathfrak{M} \models_{\mathscr{I}} \Delta$ and w_1 does not occur in Δ. Since $e(\mathscr{I}(w), \Box\alpha) \lhd \mathscr{I}(t)$ and $e(\mathscr{I}(w), \Box\alpha) = R(\mathscr{I}(w), w^\star) \to e(w^\star, \alpha)$, we get $R(\mathscr{I}(w), w^\star) \to e(w^\star, \alpha) \lhd \mathscr{I}(t)$; this proves (ii). Let $\chi \in \Phi^{\Box,\Diamond}(\Gamma, w, w_1)$; we show that $\mathfrak{M} \models_{\mathscr{I}'} \chi$, and this proves (iii). Let $\chi = R(w,w_1) \to w_1 : \beta \rhd t'$. Then, $w : \Box\beta \rhd t' \in \Delta$, hence $e(\mathscr{I}(w), \Box\beta) \rhd \mathscr{I}(t')$. It follows that $R(\mathscr{I}(w), w^\star) \to e(w^\star, \beta) \rhd \mathscr{I}(t')$, hence $\mathfrak{M} \models_{\mathscr{I}'} \chi$. The proof of $\mathfrak{M} \models_{\mathscr{I}'} \chi$ in the case $\chi = R(w,w_1) \wedge w_1 : \beta \lhd' t'$ is similar, taking into account that $w : \Diamond\beta \lhd' t' \in \Delta$. From points (i)–(iii), we conclude $\mathfrak{M} \models_{\mathscr{I}'} \Gamma_1$. □

Proposition 3.2 (Soundness). *(i) If* $\vdash_{\mathrm{GW}} \Gamma$, *then* $\mathfrak{M} \not\models \Gamma$, *for every* GW*-model* $\mathfrak{M}$.

(ii) $\vdash_{\mathrm{GW}} w : \varphi < 1$ *implies* $\varphi \in \mathbf{GW}$.

Proof. Point (i) can be proved by induction on the depth of a derivation of Γ, by exploiting Lemma 3.1. Let us assume $\varphi \notin \mathbf{GW}$. Then, there exists a GW-model $\mathfrak{M} = \langle W, R, e\rangle$ and a world $w^\star \in W$ such that $e(w^\star, \varphi) < 1$. Let $\mathscr{I}$ be an $\mathfrak{M}$-interpretation mapping w to $w^\star$; it holds that $\mathfrak{M} \models_{\mathscr{I}} w : \varphi < 1$. By (i), we get $\not\vdash_{\mathrm{GW}} w : \varphi < 1$, and this proves (ii). □

We show that $\mathscr{C}_{\mathrm{GW}}$ is *strongly terminating*, namely: there exists a well-founded relation $\prec$ such that, for every application ρ of a rule of $\mathscr{C}_{\mathrm{GW}}$, if Γ is the conclusion of ρ and Γ' any of the premises of ρ, then $\Gamma' \prec \Gamma$. Equivalently, given a finite multiset Γ and repeatedly applying the rules of $\mathscr{C}_{\mathrm{GW}}$ upwards, proof search eventually halts, no matter which strategy is used.

To prove strong termination of $\mathscr{C}_{\mathrm{GW}}$, we define the relation $\prec_{\mathrm{c}}$ between multisets of constraints. To this aim, in Fig. 8 we introduce a weight function wg on constraints and multisets of constraints Γ, together with the multisets $\Gamma[w]$ (multiset of constraints) and $||\Gamma||$ (multiset of natural numbers).

Given the finite multisets of natural numbers Θ_1 and Θ_2, we set:

$$\Theta_1 \prec_{\mathrm{m}} \Theta_2 \quad \text{iff} \quad \Theta_1 \neq \Theta_2 \ \wedge\ (\ \forall k_1 \in \Theta_1 \setminus \Theta_2.\ \exists k_2 \in \Theta_2 \setminus \Theta_1.\ k_1 < k_2\ ).$$

The relation $\prec_{\mathrm{m}}$ is a multiset order, hence $\prec_{\mathrm{m}}$ is well-founded (see [1], Th. 2.5.5 and Lemma 2.5.6). Given the multisets of constraints Γ_1 and Γ_2, we set $\Gamma_1 \prec_{\mathrm{c}} \Gamma_2$ iff $||\Gamma_1|| \prec_{\mathrm{m}} ||\Gamma_2||$.

The relation $\prec_{\mathrm{c}}$ is well-founded; in the next lemma we show that $\prec_{\mathrm{c}}$ can be used to instantiate $\prec$ in the definition of strong termination.

Lemma 3.3. *Let* ρ *be an instance of a rule of the calculus* $\mathscr{C}_{\mathrm{GW}}$, *let* Γ *be the conclusion of* ρ *and* Γ' *any of the premises of* ρ. *Then,* $\Gamma' \prec_{\mathrm{c}} \Gamma$.

Proof. We only discuss two representative cases. Let us consider the following application of rule $\to<$:

$$\frac{\overbrace{w : \alpha > w : q,\ w : \beta \leq w : q,\ w : q < t,\ \Delta_w,\ \Theta}^{\Gamma_1}}{\underbrace{w : \alpha \to \beta < t,\ \Delta_w,\ \Theta}_{\Gamma}}\ \to<$$

all the constraints in Δ_w have label w
no constraint in Θ has label w

$\Gamma[w] = \{w : \alpha \to \beta < t\} \cup \Delta_w \qquad \Gamma_1[w] = \{w : \alpha > w : q,\ w : \beta \leq w : q,\ w : q < t\} \cup \Delta_w$

$||\Gamma|| = \{\mathrm{wg}(\Gamma[w])\} \cup ||\Theta|| \qquad ||\Gamma_1|| = \{\mathrm{wg}(\Gamma_1[w])\} \cup ||\Theta||$

To prove that $||\Gamma_1|| \prec_{\mathrm{m}} ||\Gamma||$, we show that $\mathrm{wg}(\Gamma_1[w]) < \mathrm{wg}(\Gamma[w])$. The following holds:

$$\begin{array}{rcl} \mathrm{wg}(\Gamma[w]) & = & \mathrm{wg}(w:\alpha\to\beta<t)+\mathrm{wg}(\Delta_w) \;=\; \mathrm{wg}(\alpha)+\mathrm{wg}(\beta)+1+\mathrm{wg}(\Delta_w) \\ \mathrm{wg}(\Gamma_1[w]) & = & \mathrm{wg}(w:\alpha>w:q)+\mathrm{wg}(w:\beta\le w:q)+\mathrm{wg}(w:q<t)+\mathrm{wg}(\Delta_w) \\ & = & \mathrm{wg}(\alpha)+\mathrm{wg}(\beta)+0+\mathrm{wg}(\Delta_w) \end{array}$$

Since $\mathrm{wg}(\Gamma_1[w]) < \mathrm{wg}(\Gamma[w])$, we get $||\Gamma_1|| \prec_{\mathrm{m}} ||\Gamma||$, hence $\Gamma_1 \prec_{\mathrm{c}} \Gamma$.

Let us consider the following application of $\Box\lhd$:

$$\frac{\overbrace{\Delta_{w_1},\,\Delta_w,\,\Theta}^{\Gamma_1}}{\underbrace{w:\Box\alpha\lhd t,\,\Delta_w,\,\Theta}_{\Gamma}}\,\Box\lhd$$

all the constraints in Δ_w have label w
no constraint in Θ has label w
$\Delta_{w_1} \;=\; \{R(w,w_1)\to w_1:\alpha\lhd t\}\cup\Phi^{\Box,\Diamond}(\Delta_w,w,w_1)$

$$\Gamma[w]=\{w:\Box\alpha\lhd t\}\cup\Delta_w \qquad \Gamma_1[w]=\Delta_w \qquad \Gamma_1[w_1]=\Delta_{w_1}$$
$$||\Gamma|| \;=\; \{\mathrm{wg}(\Gamma[w])\}\cup||\Theta|| \qquad ||\Gamma_1|| \;=\; \{\mathrm{wg}(\Gamma_1[w]),\mathrm{wg}(\Gamma_1[w_1])\}\cup||\Theta||$$

We prove that $||\Gamma_1|| \prec_{\mathrm{m}} ||\Gamma||$. To this aim, we show that:

(i) $\mathrm{wg}(\Gamma_1[w]) < \mathrm{wg}(\Gamma[w])$.

(ii) $\mathrm{wg}(\Gamma_1[w_1]) < \mathrm{wg}(\Gamma[w])$.

The proof of point (i) is immediate since $\Gamma[w]=\{w:\Box\alpha\lhd t\}\cup\Gamma_1[w]$. We prove (ii). We have:

$$\mathrm{wg}(\Gamma[w]) \;=\; \mathrm{wg}(w:\Box\alpha\lhd t)+\mathrm{wg}(\Delta_w) \;=\; \mathrm{wg}(\alpha)+2+\mathrm{wg}(\Delta_w)$$
$$\mathrm{wg}(\Gamma_1[w_1]) \;=\; \mathrm{wg}(R(w,w_1)\to w_1:\alpha)+\mathrm{wg}(\Phi^{\Box,\Diamond}(\Delta_w,w,w_1)) \;=\; \mathrm{wg}(\alpha)+1+\mathrm{wg}(\Phi^{\Box,\Diamond}(\Delta_w,w,w_1))$$

To conclude the proof of (ii), we show that:

(iii) $\mathrm{wg}(\Phi^{\Box,\Diamond}(\Delta_w,w,w_1)) < \mathrm{wg}(\Delta_w)$.

Let $\chi_1\in\Phi^{\Box,\Diamond}(\Delta_w,w,w_1)$. There exists $\chi\in\Delta_w$ such that one of the two following conditions holds:

(a) $\chi = w:\Box\beta\rhd' t'$ and $\chi_1 = R(w,w_1)\to w_1:\beta\rhd' t'$;

(b) $\chi = w:\Diamond\beta\lhd' t'$ and $\chi_1 = R(w,w_1)\wedge w_1:\beta\lhd' t'$.

In both cases, it holds that $\mathrm{wg}(\chi_1)=\mathrm{wg}(\beta)+1$ and $\mathrm{wg}(\chi)=\mathrm{wg}(\beta)+2$, hence $\mathrm{wg}(\chi_1)<\mathrm{wg}(\chi)$; this proves (iii). From (i) and (ii) it follows that $||\Gamma_1|| \prec_{\mathrm{m}} ||\Gamma||$, hence $\Gamma_1\prec_{\mathrm{c}}\Gamma$. □

As an immediate consequence of Lemma 3.3, we get:

Proposition 3.4. *The calculus $\mathscr{C}_{\mathrm{GW}}$ is strongly terminating.*

Consequently, any backward proof search strategy for $\mathscr{C}_{\mathrm{GW}}$ terminates. Following [8, 12], we focus on strategies where failure is certified by countermodels, i.e.: if proof search for a multiset of constraints Γ fails, a GW-model $\mathfrak{M}$ such that $\mathfrak{M}\models\Gamma$ can be built; we call $\mathfrak{M}$ a *countermodel* for Γ, since $\mathfrak{M}$ ascertains that Γ is not provable in $\mathscr{C}_{\mathrm{GW}}$ (by its soundness). Let Γ be a multiset of constraints:

- Γ is *reduced* iff no rule of $\mathscr{C}_{\mathrm{GW}}$ can be backward applied to Γ; thus, every non-atomic constraint in Γ has the form $w:\Box\alpha\rhd t$ or $w:\Diamond\alpha\lhd t$.

$$\mathrm{wg}(u) = \begin{cases} 0 & \text{if } u \text{ is an atomic c-term} \\ |\varphi| & \text{if } u = w : \varphi \\ |\varphi| + 1 & \text{if } u = R(w,w') \star w' : \varphi \\ & \quad \text{with } \star \in \{\wedge, \rightarrow\} \end{cases} \qquad |\varphi| = \begin{cases} 0 & \text{if } \varphi \in \mathscr{V} \cup \{\bot\} \\ |\alpha| + |\beta| + 1 & \text{if } \varphi = \alpha \star \beta \\ & \quad \text{with } \star \in \{\wedge, \vee, \rightarrow\} \\ |\alpha| + 2 & \text{if } \varphi = \Box\alpha \text{ or } \varphi = \Diamond\alpha \end{cases}$$

$$\mathrm{wg}(u \blacktriangledown t) = \mathrm{wg}(u) \qquad \mathrm{wg}(\Gamma) = \textstyle\sum_{\chi \in \Gamma} \mathrm{wg}(\chi)$$

$$\Gamma[w] = \{\, u \blacktriangledown t \in \Gamma \mid \mathrm{label}(u) = w \,\} \qquad ||\Gamma|| = \{\, \mathrm{wg}(\Gamma[w]) \mid w \text{ occurs in } \Gamma \,\}$$

Figure 8: Weight function wg on constraints and the multisets $\Gamma[w]$, $||\Gamma||$.

- Γ is *plain* iff only a modal rule can be backward applied to Γ; thus, every non-atomic constraint in Γ has the form $w : \Box\alpha \blacktriangledown t$ or $w : \Diamond\alpha \blacktriangledown t$.

An application of a modal rule is plain iff its conclusion is plain. By $\Gamma_k \mapsto_\chi \Gamma_{k+1}$ we mean that Γ_{k+1} is a premise of an application of a rule ρ of $\mathscr{C}_{\mathrm{GW}}$ having conclusion Γ_k and main constraint χ, with $\chi \in \Gamma_k$. A branch $\mathscr{B}$ is a sequence $\langle \Gamma_0, \ldots, \Gamma_n \rangle$ such that, for every $0 \leq k < n$, there is χ_k such that $\Gamma_k \mapsto_{\chi_k} \Gamma_{k+1}$; thus, $\mathscr{B}$ represents a branch of a tree of $\mathscr{C}_{\mathrm{GW}}$. The branch $\mathscr{B}$ is *saturated* iff the following holds:

- Γ_0 is a finite set of constraints of the form $w : \varphi \blacktriangledown t$ and Γ_n is reduced;
- every application of a modal rule in $\mathscr{B}$ is plain.

We show how to construct a countermodel for Γ_0 from Γ_n. Let Γ_{at} be a finite consistent set of atomic constraints and let σ be a solution to Γ_{at}. By $\mathrm{Mod}(\Gamma_{\mathrm{at}}, \sigma)$ we denote the G-model $\mathfrak{M} = \langle W, R, e \rangle$ such that W is the set of labels occurring in Γ_{at} and:

$$R(w,w') = \begin{cases} \sigma(R(w,w')) & \text{if } R(w,w') \text{ occurs in } \Gamma_{\mathrm{at}} \\ 0 & \text{otherwise} \end{cases} \qquad e(w,p) = \begin{cases} \sigma(w : p) & \text{if } w : p \text{ occurs in } \Gamma_{\mathrm{at}} \\ 0 & \text{otherwise} \end{cases}$$

We remark that $\mathfrak{M}$ is a discrete GW-model. A $\mathrm{Mod}(\Gamma_{\mathrm{at}}, \sigma)$-interpretation $\mathscr{I}$ is *canonical* iff $\mathscr{I}(w) = w$ for every label w occurring in Γ_{at}. In Lemma 3.6 we show that if $\mathscr{B} = \langle \Gamma_0, \ldots, \Gamma_n \rangle$ is saturated and σ is a solution to $\mathrm{At}^+(\Gamma_n)$, then $\mathrm{Mod}(\mathrm{At}^+(\Gamma_n), \sigma)$ is a countermodel for Γ_0.

Example 3.5. Let $\varphi = \Box(p \vee q) \rightarrow (\Box p \vee \Diamond q)$ be an instance of (Cr). The tree shown in Fig. 9 corresponds to a saturated branch $\mathscr{B} = \langle \Gamma_0, \ldots, \Gamma_7 \rangle$, where $\Gamma_0 = \{w_0 : \varphi < 1\}$ is the root and Γ_7 is the top multiset. When the applied rule has two premises, the annotation (l) (left) or (r) (right) specifies the selected one; in every rule application, the main constraint is underlined.

The set $\mathrm{At}^+(\Gamma_7)$ consists of the following atomic constraints:

$$w_1 : p < R(w_0,w_1),\ \ w_1 : p \leq c,\ \ w_1 : q \geq R(w_0,w_1),\ 1 > c,\ \ R(w_0,w_1) \leq c,\ \ c < 1$$

The set $\mathrm{At}^+(\Gamma_7)$ is consistent; a solution to $\mathrm{At}^+(\Gamma_7)$ is any substitution σ over $\mathrm{At}^+(\Gamma_7)$ such that

$$\sigma(w_1 : p) < \sigma(R(w_0,w_1)) \leq \sigma(c) < 1 \qquad \sigma(R(w_0,w_1)) \leq \sigma(w_1 : q)$$

For every solution σ, the GW-model $\mathfrak{M} = \mathrm{Mod}(\mathrm{At}^+(\Gamma_7), \sigma)$ satisfies $\mathfrak{M} \models_{\mathscr{I}} w_0 : \varphi < 1$, where $\mathscr{I}$ is a canonical $\mathfrak{M}$-interpretation; as a consequence, $\mathfrak{M}$ is a countermodel for φ, witnessing that $\varphi \notin \mathbf{GW}$. Note that the worlds of $\mathfrak{M}$ are w_0 and w_1 and, as expected, $\mathfrak{M}$ is not crisp, since $0 < \sigma(R(w_0,w_1)) < 1$. A concrete solution $\sigma^\star$ is obtained by setting $\sigma^\star(w_1 : p) = 0.5$ and $\sigma^\star(R(w_0,w_1)) = \sigma^\star(c) = \sigma^\star(w_1 : q) = 0.6$. The model $\mathrm{Mod}(\mathrm{At}^+(\Gamma_7), \sigma^\star)$ is essentially the same as the one displayed in Fig. 3. $\Diamond$

$$c = w_0 : q_0 \qquad \Delta \;=\; \{\, w_0 : \Box(p \vee q) > c,\;\; w_0 : \Diamond q \leq c,\;\; c < 1 \,\}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{w_1 : p < R(w_0,w_1),\;\; w_1 : p \leq c,\;\; w_1 : q \geq R(w_0,w_1),\;\; 1 > c,\;\; R(w_0,w_1) \leq c,\;\; \Delta}{w_1 : p < R(w_0,w_1),\;\; w_1 : p \leq c,\;\; w_1 : q \geq R(w_0,w_1),\;\; 1 > c,\;\; \underline{R(w_0,w_1) \wedge w_1 : q \leq c},\;\; \Delta}\, R\wedge\lhd(l)}{w_1 : p < R(w_0,w_1),\;\; w_1 : p \leq c,\;\; \underline{w_1 : p \vee q \geq R(w_0,w_1)},\;\; 1 > c,\;\; R(w_0,w_1) \wedge w_1 : q \leq c,\;\; \Delta}\, \vee\rhd(r)}{w_1 : p < R(w_0,w_1),\;\; w_1 : p \leq c,\;\; \underline{R(w_0,w_1) \to w_1 : p \vee q > c},\;\; R(w_0,w_1) \wedge w_1 : q \leq c,\;\; \Delta}\, R\to\rhd(l)}{\underline{R(w_0,w_1) \to w_1 : p \leq c},\;\; R(w_0,w_1) \to w_1 : p \vee q > c,\;\; R(w_0,w_1) \wedge w_1 : q \leq c,\;\; \Delta}\, R\to\leq(r)}{w_0 : \Box(p \vee q) > c,\;\; \underline{w_0 : \Box p \leq c},\;\; w_0 : \Diamond q \leq c,\;\; c < 1}\, \Box\lhd}{w_0 : \Box(p \vee q) > c,\;\; \underline{w_0 : (\Box p \vee \Diamond q) \leq c},\;\; c < 1}\, \vee\lhd}{\underline{w_0 : \Box(p \vee q) \to (\Box p \vee \Diamond q) < 1}}\, \to<$$

Figure 9: A saturated branch.

Lemma 3.6. *Let $\mathscr{B} = \langle \Gamma_0, \ldots, \Gamma_n \rangle$ be a saturated branch and σ a solution to $\mathrm{At}^+(\Gamma_n)$. Let $\mathfrak{M} = \mathrm{Mod}(\mathrm{At}^+(\Gamma_n), \sigma)$, $\mathscr{I}$ a canonical $\mathfrak{M}$-interpretation and $\chi \in \bigcup_{k\in\{0,\ldots,n\}} \Gamma_k$. Then, $\mathfrak{M} \models_{\mathscr{I}} \chi$.*

Proof. Since $\mathfrak{M} = \mathrm{Mod}(\mathrm{At}^+(\Gamma_n), \sigma)$, with σ is a solution to $\mathrm{At}^+(\Gamma_n)$, it holds that:

(1) $\mathfrak{M} \models_{\mathscr{I}} \mathrm{At}^+(\Gamma_n)$.

We prove $\mathfrak{M} \models_{\mathscr{I}} \chi$ by induction on $\mathrm{wg}(\chi)$; we assume $\mathfrak{M} = \langle W, R, e \rangle$. Let $\mathrm{wg}(\chi) = 0$; then χ is atomic, hence $\chi \in \Gamma_n$, and $\mathfrak{M} \models_{\mathscr{I}} \chi$ by (1).

Let $\mathrm{wg}(\chi) > 0$ and let us assume that $\chi \notin \Gamma_n$. Then, there exists $k \geq 0$ such that $\Gamma_k \mapsto_\chi \Gamma_{k+1}$. We proceed by a case analysis on the structure of χ.

Let $\chi = w : \alpha \wedge \beta \rhd t$. Since $\Gamma_k \mapsto_\chi \Gamma_{k+1}$, the applied rule is $\wedge\rhd$, hence $w : \alpha \rhd t \in \Gamma_{k+1}$ and $w : \beta \rhd t \in \Gamma_{k+1}$. By the induction hypothesis $\mathfrak{M} \models_{\mathscr{I}} w : \alpha \rhd t$ and $\mathfrak{M} \models_{\mathscr{I}} w : \beta \rhd t$, and this implies $\mathfrak{M} \models_{\mathscr{I}} w : \alpha \wedge \beta \rhd t$. The case $\chi = w : \alpha \vee \beta \lhd t$ is similar.

Let $\chi = w : \alpha \wedge \beta \lhd t$. Since $\Gamma_k \mapsto_\chi \Gamma_{k+1}$, the applied rule is $\wedge\lhd$, hence $w : \alpha \lhd t \in \Gamma_{k+1}$ or $w : \beta \lhd t \in \Gamma_{k+1}$. By the induction hypothesis $\mathfrak{M} \models_{\mathscr{I}} w : \alpha \lhd t$ or $\mathfrak{M} \models_{\mathscr{I}} w : \beta \lhd t$, hence $\mathfrak{M} \models_{\mathscr{I}} w : \alpha \wedge \beta \lhd t$. The case $\chi = w : \alpha \vee \beta \rhd t$ is similar.

Let $\chi = w : \alpha \to \beta < t$; we only consider the case $\beta \notin \mathscr{V} \cup \{\bot\}$. Since $\Gamma_k \mapsto_\chi \Gamma_{k+1}$, the applied rule is $\to<$, hence:

(A) $\{\, w : \alpha > w : q,\; w : \beta \leq w : q,\; w : q < t \,\} \subseteq \Gamma_{k+1}$.

By the induction hypothesis, for every constraint χ' displayed in (A), $\mathfrak{M} \models_{\mathscr{I}} \chi'$, hence:

$$e(w,\alpha) > e(w,q) \qquad e(w,\beta) \leq e(w,q) \qquad e(w,q) < \mathscr{I}(t)$$

Since $e(w,\alpha) > e(w,\beta)$, we get $e(w,\alpha \to \beta) = e(w,\beta)$, hence $e(w,\alpha \to \beta) < \mathscr{I}(t)$, which implies $\mathfrak{M} \models_{\mathscr{I}} w : \alpha \to \beta < t$.

Let $\chi = w : \alpha \to \beta \leq t$; we only consider the case $\beta \notin \mathscr{V} \cup \{\bot\}$. Since $\Gamma_k \mapsto_\chi \Gamma_{k+1}$, the applied rule is $\to\leq$, hence one of the following two subcases holds:

(B1) $t \geq 1 \in \Gamma_{k+1}$;

(B2) $\{\, w : \alpha > w : q,\; w : \beta \leq w : q,\; w : q \leq t \,\} \subseteq \Gamma_{k+1}$.

Let us assume that (B1) holds. By the induction hypothesis, we get $\mathfrak{M} \models_{\mathscr{I}} t \geq 1$, and this implies $\mathfrak{M} \models_{\mathscr{I}} w : \alpha \to \beta \leq t$. Let us assume that (B2) holds. By the induction hypothesis, for every constraint χ' in (B2), $\mathfrak{M} \models_{\mathscr{I}} \chi'$, hence:

$$e(w,\alpha) > e(w,q) \qquad e(w,\beta) \leq e(w,q) \qquad e(w,q) \leq \mathscr{I}(t)$$

Since $e(w,\alpha) > e(w,\beta)$, we get $e(w,\alpha \to \beta) = e(w,\beta)$, hence $e(w,\alpha \to \beta) \leq \mathscr{I}(t)$, which implies $\mathfrak{M} \models_{\mathscr{I}} w : \alpha \to \beta \leq t$.

Let $\chi = w : \alpha \to \beta \rhd t$; we only consider the case $\beta \notin \mathscr{V} \cup \{\bot\}$. Since $\Gamma_k \mapsto_\chi \Gamma_{k+1}$, the applied rule is $\to \rhd$, hence one of the following two subcases holds:

(C1) $\{ w : \alpha \leq w : q, w : \beta \geq w : q, 1 \rhd t \} \subseteq \Gamma_{k+1}$;

(C2) $w : \beta \rhd t \in \Gamma_{k+1}$.

Assume (C1). By the induction hypothesis, for every constraint χ' in (C1), $\mathfrak{M} \models_{\mathscr{I}} \chi'$, hence:

$$e(w,\alpha) \leq e(w,q) \qquad e(w,\beta) \geq e(w,q) \qquad 1 \rhd \mathscr{I}(t)$$

Since $e(w,\alpha) \leq e(w,\beta)$, we get $e(w,\alpha \to \beta) = 1$, hence $e(w,\alpha \to \beta) \rhd \mathscr{I}(t)$, which implies $\mathfrak{M} \models_{\mathscr{I}} \chi$. In case (C2), by the induction hypothesis we get $\mathfrak{M} \models_{\mathscr{I}} w : \beta \rhd t$, namely $e(w,\beta) \rhd \mathscr{I}(t)$. Since $e(w,\alpha \to \beta) \geq e(w,\beta)$, we get $e(w,\alpha \to \beta) \rhd \mathscr{I}(t)$, hence $\mathfrak{M} \models_{\mathscr{I}} w : \alpha \to \beta \rhd t$. The cases concerning R-constraints can be proved as the corresponding propositional cases.

Let $\chi = w : \Box\alpha \lhd t$. Since $\Gamma_k \mapsto_\chi \Gamma_{k+1}$, the applied rule is $\Box\lhd$, hence:

(D) $R(w,w_1) \to w_1 : \alpha \lhd t \in \Gamma_{k+1}$.

Let r_1 be the value of $R(w,w_1) \to e(w_1 : \alpha)$. By the induction hypothesis $\mathfrak{M} \models_{\mathscr{I}} R(w,w_1) \to w_1 : \alpha \lhd t$, hence $r_1 \lhd \mathscr{I}(t)$. Since $e(w,\Box\alpha) \leq r_1$, we get $e(w,\Box\alpha) \lhd \mathscr{I}(t)$, and this implies $\mathfrak{M} \models_{\mathscr{I}} w : \Box\alpha \lhd t$. The case $\chi = w : \Diamond\alpha \rhd t$ is similar.

It remains to consider the cases where $\mathrm{wg}(\chi) > 0$ and $\chi \in \Gamma_n$, thus $\chi = w : \Box\alpha \rhd t$ or $\chi = w : \Diamond\alpha \lhd t$. Let $\chi = w : \Box\alpha \rhd t$; we show that:

(E) $\mathfrak{M} \models_{\mathscr{I}} R(w,w') \to w' : \alpha \rhd t$, for every $w' \in W$.

Let $w' \in W$; we show that $\mathfrak{M} \models_{\mathscr{I}} R(w,w') \to w' : \alpha \rhd t$. Let us assume that $R(w,w')$ does not occur in Γ_n. By definition of $\mathfrak{M}$, the value of $R(w,w')$ is 0, hence we have to show that $\mathfrak{M} \models_{\mathscr{I}} 1 \rhd t$. The case where $\rhd$ is $\geq$ is immediate. Let $\rhd$ be $>$. Since $\chi \in \Gamma_n$, we get $1 > t \in \mathrm{At}^+(\Gamma_n)$, thus $\mathfrak{M} \models_{\mathscr{I}} 1 > t$ by (1). Let us assume that $R(w,w')$ occurs in Γ_n. Then, there exists $k > 0$ such that $R(w,w')$ is introduced in Γ_k by an application of a modal rule ρ, namely:

- $\Gamma_{k-1} \mapsto_{\chi'} \Gamma_k$, with $\chi' = w : \Box\varphi \lhd' t'$ or $\chi' = w : \Diamond\varphi \rhd' t'$, and the new label introduced by ρ is w'.

Since $\mathscr{B}$ is saturated, Γ_{k-1} is plain, hence the non-atomic constraints in Γ_{k-1} are modal. As a consequence, no rule applied in the sub-branch $\langle \Gamma_{k-1}, \ldots, \Gamma_n \rangle$ can introduce in the premise new constraints $u \blacktriangledown t'$ such that u has label w; this implies that $\chi \in \Gamma_{k-1}$. By definition of ρ, we get $\Phi^{\Box,\Diamond}(\Gamma_{k-1}, w, w') \subseteq \Gamma_k$, hence $R(w,w') \to w' : \alpha \rhd t \in \Gamma_k$. By the induction hypothesis, we get $\mathfrak{M} \models_{\mathscr{I}} R(w,w') \to w' : \alpha \rhd t$, and this concludes the proof of (E). Let $e(w,\Box\alpha) = r$. By the witnessing condition, there exists $w^\star \in W$ such that the value of $R(w,w^\star) \to e(w^\star,\alpha)$ is r. By (E), $r \rhd \mathscr{I}(t)$, namely $e(w,\Box\alpha) \rhd \mathscr{I}(t)$; we conclude $\mathfrak{M} \models_{\mathscr{I}} w : \Box\alpha \rhd t$. The case $\chi = w : \Diamond\alpha \lhd t$ is similar. □

As a consequence, we get:

Proposition 3.7. *Let $\mathscr{B} = \langle \Gamma_0, \ldots, \Gamma_n \rangle$ be a saturated branch.*

(i) For every solution σ *to* $\mathrm{At}^+(\Gamma_n)$, $\mathrm{Mod}(\mathrm{At}^+(\Gamma_n),\sigma)$ *is a discrete countermodel for* Γ_0.

(ii) If $\Gamma_0 = \{w : \varphi < 1\}$, *then* $\varphi \notin \mathbf{GW}$.

Let BS be a backward proof search strategy for $\mathscr{C}_{\mathrm{GW}}$; we say that BS is *plain* iff all the modal rule applications performed by BS are plain.

Lemma 3.8. *Let* BS *be a plain proof search strategy for* $\mathscr{C}_{\mathrm{GW}}$ *and let* Γ_0 *be a finite multiset of constraints of the form* $w : \varphi \blacktriangledown t$. *If* BS *fails to prove* Γ_0, *then a discrete countermodel for* Γ_0 *can be built.*

Proof. Assume that BS fails. By tracing the computation, we can build an open branch $\mathscr{B} = \langle \Gamma_0, \ldots, \Gamma_n \rangle$. Since BS is plain, the branch $\mathscr{B}$ is saturated. Let σ be a solution to $\mathrm{At}^+(\Gamma_n)$; by Prop. 3.7(i), we get that $\mathrm{Mod}(\mathrm{At}^+(\Gamma_n),\sigma)$ is a discrete countermodel for Γ_0. □

To sum up, we get the completeness of $\mathscr{C}_{\mathrm{GW}}$ and the finite model property for **GW**.

Proposition 3.9 (Completeness)**.** *Let* Γ *be a finite multiset of constraints of the form* $w : \varphi \blacktriangledown t$.

(i) $\vdash_{\mathrm{GW}} \Gamma$ *iff, for every* GW*-model* $\mathfrak{M}$, $\mathfrak{M} \not\models \Gamma$.

(ii) $\varphi \in \mathbf{GW}$ *iff* $\vdash_{\mathrm{GW}} w : \varphi < 1$; *moreover, if* $\varphi \notin \mathbf{GW}$, *there exists a discrete countermodel for* φ.

Let $\varphi \notin \mathbf{GW}$ and let $\mathfrak{M}$ be the discrete countermodel extracted from an opennn branch with root $w_0 : \varphi < 1$. One can easily show that the depth of $\mathfrak{M}$ is bounded by $|\varphi|$ and that every world in $\mathfrak{M}$ has at most $|\varphi|$ R-successors. Consequently, the size of $\mathfrak{M}$ is $O(|\varphi|^{|\varphi|})$; moreover, we can define formulas φ having countermodels of size $2^{|\varphi|}$ (e.g., consider the translation φ^* of a **K**-formula φ such that φ has a countermodel of exponential size). Note that $\mathfrak{M}$ cannot be constructed incrementally, one branch at a time, since the constraints generated during the expansion of a branch have global validity and must be preserved throughout the entire construction. Nevertheless, we can define a PSPACE algorithm to check if such a countermodel exists without effectively constructing it; this is discussed in the proof of the next proposition.

Proposition 3.10. *The validity problem for* **GW** *is PSPACE-complete.*

Proof. Since the modal logic **K** can be embedded into **GW** (Prop. 2.2) and **K** is PSPACE-complete [14], the validity problem for **GW** is PSPACE-hard. It remains to prove that **GW**-validity can be checked in PSPACE. To this aim, we sketch a PSPACE procedure to decide the membership of a formula to **GW**. Henceforth, we assume that the formula to be checked is φ. As discussed above, if $\varphi \notin \mathbf{GW}$, then there exists a tree-like countermodel $\mathfrak{M} = \langle W, R, e \rangle$ for φ such that the number of worlds is $O(|\varphi|^{|\varphi|})$. We can also bound the number N of values needed to define R and e: in the worst case, the values $e(w,p)$ and $R(w,w')$ must all be pairwise distinct, for every world w, w' in $\mathfrak{M}$ and every propositional variable p occurring in φ; accordingly, N is in $O(|\varphi|^{|\varphi|})$. We set $\mathscr{Q} = \{0\} \cup \{\ 1/k \mid 1 \leq k \leq N\}$; note that the elements of $\mathscr{Q}$ can be generated in space $O(|\varphi|)$. By the above discussion, if $\varphi \notin \mathbf{GW}$ there exists a countermodel $\mathfrak{M}$ such that the images of R and e are contained in $\mathscr{Q}$. To define the procedure, we need a global map Φ that specifies the intended values of the subformulas of φ at each world of the model under construction. More precisely, given a world name w, $\Phi(w)$ is a set of assignments of the form $e(w,\alpha) = r$, where α is a subformula of φ and $r \in \mathscr{Q}$. We remark that $\Phi(w)$ can be stored in space $O(|\varphi|)$; moreover, to implement the search procedure, we do no not need to store the whole map Φ, but at most $O(|\varphi|)$ entries of Φ. A GW-model $\mathfrak{M} = \langle W, R, e \rangle$ complies with Φ iff, for every w in W and every assignment $e(w,\alpha) = r$ stored in $\Phi(w)$, the value of $e(w,\alpha)$ (in $\mathfrak{M}$) is r. The decision procedure is implemented by the recursive procedure SEARCH having the following specification:

- Let w be a world name and let us assume that $\Phi(w) \neq \emptyset$. The procedure SEARCH(w) returns SUCCESS if a GW-model complying with Φ can be built, FAIL otherwise.

The procedure, at some points, requires the selection of a value from $\mathscr{Q}$, possibly satisfying some conditions: we call this a $\mathscr{Q}$-choice. $\mathscr{Q}$-choices can be revoked and redone; such backtrack steps can be implemented in space $O(|\varphi|)$. The computation of SEARCH(w) consists of two main steps.

(Step 1) We apply propositional saturation reductions to $\Phi(w)$ as long as possible. At each step, an assignment $e(w,\alpha \star \beta) = r$ (with $\star \in \{\wedge, \vee, \rightarrow\}$) in $\Phi(w)$ is removed and replaced by the assignments $e(w,\alpha) = r_1$ and $e(w,\beta) = r_2$ required to justifies the value r, where the values r_1 and r_2 are selected with $\mathscr{Q}$-choices. For instance, let us assume that the assignment at hand is $e(w,\alpha \rightarrow \beta) = r$.

- If $r = 1$, we must have $e(w,\alpha) \leq e(w,\beta)$; the values r_1 and r_2 are selected with $\mathscr{Q}$-choices so that $r_1 \leq r_2$; in $\Phi(w)$, $e(w,\alpha \rightarrow \beta) = r$ is replaced with $e(w,\alpha) = r_1$ and $e(w,\beta) = r_2$.
- If $r < 1$, a value r_1 is selected with a $\mathscr{Q}$-choice so that $r_1 > r$; in $\Phi(w)$, $e(w,\alpha \rightarrow \beta) = r$ is replaced with $e(w,\alpha) = r_1$ and $e(w,\beta) = r$ (thus, r_2=r).

At the end of (Step 1), the formulas occurring in $\Phi(w)$ are either atomic or modal.

- If $\Phi(w)$ contains two evaluations $e(w,\alpha) = r_1$ and $e(w,\alpha) = r_2$ such that $r_1 \neq r_2$, a backtrack step is needed. If no new $\mathscr{Q}$-choice is possible, the call SEARCH(w) returns FAIL.
- Otherwise, if $\Phi(w)$ contains only atomic constraints, the call SEARCH(w) returns SUCCESS.
- Otherwise, $\Phi(w)$ contains at least a modal formula and the computation continues with (Step 2).

(Step 2) Let $e(w,\#_1\alpha_1) = r_1, \ldots, e(w,\#_n\alpha_n) = r_n$ (with $\#_k \in \{\Box, \Diamond\}$) be all the assignments in $\Phi(w)$ involving modal formulas. We start a loop indexed by k ranging over $\{1,\ldots,n\}$. Let k be the current iteration and let us assume that the assignment of index k has the form $e(w,\Box\alpha_k) = r_k$ (the case $e(w,\Diamond\alpha_k) = r_k$ is similar). We generate a new world w_k and select with a $\mathscr{Q}$-choice a value r'_k for $R(w,w_k)$ and a value r''_k for $e(w_k,\alpha_k)$ such that $r'_k \rightarrow r''_k = r_k$. The world w_k is intended to represent the world witnessing that $e(w,\Box\alpha_k) = r_k$. We set $\Phi(w_k) = \{e(w_k,\alpha_k) = r''_k\}$. We expand $\Phi(w_k)$ by adding the assignments required to account for the assignments $e(w,\#_j\alpha_j) = r_j$ occurring in $\Phi(w)$, where $j \neq k$. More precisely:

- for every $e(w,\Box\alpha_j) = r_j$ in $\Phi(w)$ ($j \neq k$), we select with a $\mathscr{Q}$-choice a value r'_j such that $r'_k \rightarrow r'_j \geq r_j$ and we add $e(w_k,\alpha_j) = r'_j$ to $\Phi(w_k)$;
- for every $e(w,\Diamond\alpha_j) = r_j$ in Φ ($j \neq k$), we select with a $\mathscr{Q}$-choice a value r'_j such that $r'_k \wedge r'_j \leq r_j$ and we add $e(w_k,\alpha_j) = r'_j$ to $\Phi(w_k)$.

We compute recursively SEARCH(w_k); notice that the modal operators occurring in $\Phi(w_k)$ are fewer than the modal operators in $\Phi(w)$, thus the recursion is terminating. We also stress that there are at most $|\varphi|$ nested recursive calls. Let us assume that SEARCH(w_k) returns SUCCESS. If $k = n$ (last iteration), the call SEARCH(w) returns SUCCESS. Otherwise the computation continues with next iteration $k+1$. We point out that the entries of the map Φ exploited in the computation of SEARCH(w_k) are no longer required, so they can be removed from the memory. Let us assume that SEARCH(w_k) returns FAIL. Then, a backtrack step is needed. If no new $\mathscr{Q}$-choice is possible, the call SEARCH(w) returns FAIL.

The procedure can be implemented so that SEARCH(w) is computed in space $O(|\varphi|)$.

We can exploit SEARCH to decide the validity of φ in **GW**. We initialize Φ by setting $\Phi(w_0) = \{e(w_0, \varphi) = r_0\}$, where r_0 is any value in $\mathscr{Q}$ different from 1. If SEARCH(w_0) returns SUCCESS, then a model $\mathfrak{M}$ containing a world w_0 such that $e(w_0, \varphi) = r_0$ can be built; since $r_0 < 1$, we conclude $\varphi \notin \mathbf{GW}$. If SEARCH$(w_0)$ returns FAIL, for every possible choice of r_0, it follows that no GW-countermodel for φ can be built; accordingly, $\varphi \in \mathbf{GW}$. □

4 Conclusions and Future Work

We have implemented a proof-search procedure for the calculus $\mathscr{C}_{\mathrm{GW}}$ that supports countermodel extraction. The prover, called `gwref` [10], performs a standard backward depth-first proof search, relying on the JTabWb engine [9]. The consistency of atomic constraints is verified using the Choco-solver Java library [16]. We emphasize that the calculus $\mathscr{C}_{\mathrm{GW}}$ and the associated proof-search strategy can be smoothly adapted to treat the logic $\mathbf{GW}^{\mathrm{c}}$ [11], characterized by crisp GW-models, by refining the notion of consistency for a set of atomic constraints Γ_{at}: a solution σ to Γ_{at} must satisfy $\sigma(R(w, w')) \in \{0, 1\}$ for every pair of labels w, w'. To the best of our knowledge, the only other prover available for modal fuzzy logics is mNiBLoS [20], an SMT-based solver designed for continuous t-norm-based logics, including Gödel–Dummett Logic. However, neither **GW** nor $\mathbf{GW}^{\mathrm{c}}$ are supported by mNiBLoS.

The results presented in this paper are highly relevant to the study of intermediate predicate logics. It is well known that modal logics can be used to characterize relevant fragments of intermediate first-order logics, particularly certain one-variable fragments. From this perspective, the witnessed semantics, together with the consequent finite model property established here, point to deep connections with proof-theoretic properties of intermediate predicate logics, such as Herbrand's Theorem, interpolation, and alternative forms of Skolemization investigated in [2] for broad classes of intermediate intuitionistic predicate logics with the finite model property. Motivated by these connections, and with the aim of situating **GW** within the broader landscape of intuitionistic modal logics, we plan to develop a birelational Kripke semantics for **GW**, extending to this setting the approach introduced for $\mathbf{GW}^{\mathrm{c}}$ in [11].

We also plan to extend the procedure to other witnessed Gödel modal logics that impose additional constraints on the accessibility relation (e.g., reflexivity, transitivity, seriality). In contrast, treating Gödel modal logics with non-witnessed models seems to be considerably more challenging.

Acknowledgments

This project has received funding from the European Union's Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie grant agreement No 101007627 (MOSAIC project) and from the European Union's Horizon Europe research and innovation programme under the Marie Skłodowska-Curie grant agreement No. 101299559 (SemPER project).

Camillo Fiorentini is member of the *Gruppo Nazionale Calcolo Scientifico - Istituto Nazionale di Alta Matematica (GNCS-INdAM)*. Mauro Ferrari and Paolo Giardini are members of the *Gruppo Nazionale per le Strutture Algebriche, Geometriche e le loro Applicazioni - Istituto Nazionale di Alta Matematica (GNSAGA-INdAM)*.

The last author would like to remark that this work was completed despite the lack of support from the scientific funding agencies of Argentina.

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